

Lecture 5

Measures of Location and Position

Review

1 A distribution has two clear peaks. What does that usually tell us?

The sample mixes two distinct sub-populations, picked out by some grouping variable.

2 What makes a distribution skewed?

A hard limit on the values – data pile up against the limit and trail away from it.

3 Name the two very different things an outlier can be.

A genuinely unusual observation, or a measurement or recording error.

4 Which plot(s) are best for large sample sizes, and which one(s) keep the actual data values?

The histogram for large n . The stem and leaf plot keeps the original values. Dot plot also – when applied to discrete data.

Review: ten video games

X = Metacritic critic score, for a random sample of $n = 10$ best-selling video games (circa 2000's).

$x = \{ 68, 72, 76, 76, 76, 83, 85, 87, 92, 95 \}$

Game	Platform	Score
Medal of Honor: Rising Sun	PS2	68
Mario Party DS	DS	72
Destiny	PS4	76
Pokémon Emerald Version	GBA	76
Wii Sports	Wii	76
Pokémon Platinum Version	DS	83
FIFA 17	PS4	85
Dragon Quest IX: Sentinels of the Starry Skies	DS	87
Forza Motorsport 3	X360	92
Grand Theft Auto: San Andreas	PS2	95

Critic score	Frequency	Relative Frequency
60 to under 70	1	0.1
70 to under 80	4	0.4
80 to under 90	3	0.3
90 to under 100	2	0.2
Total	10	1.0

Four bins of width 10. Most of these games score in the 70s, with a few running up into the 90s.

Measures of Central Tendency: the Mean

- The (arithmetic) **mean** is the average value of a set of observations
it measures the center of mass of a distribution (the balancing point)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n}$$

- The population mean is also called the **expected value**, written $E(X)$ or μ
 - the long-run average of the variable; the sample mean is our estimate of it
- the mean is usually not equal to any of the values observed in the sample
- The mean is highly influenced by **outliers** - observations that take on extreme values relative to the distribution

Practice: Calculate The Mean

- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

Alternative formulas for the mean

- We can also express the mean in terms of the frequency F or the relative frequency RF

$$\bar{x} = \frac{1}{n} \sum_x x F(x) \quad \text{or} \quad \bar{x} = \sum_x x RF(x)$$

Where the sum is over all distinct values of the variable x

Example: Computing the mean from a frequency table

$X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

X	Freq.	Rel. Freq
1	1	0.125
3	1	0.125
5	2	0.250
6	1	0.125
7	2	0.250
8	1	0.125

Measures of Central Tendency: the Median

- The **median** is the middle value of a set of observations

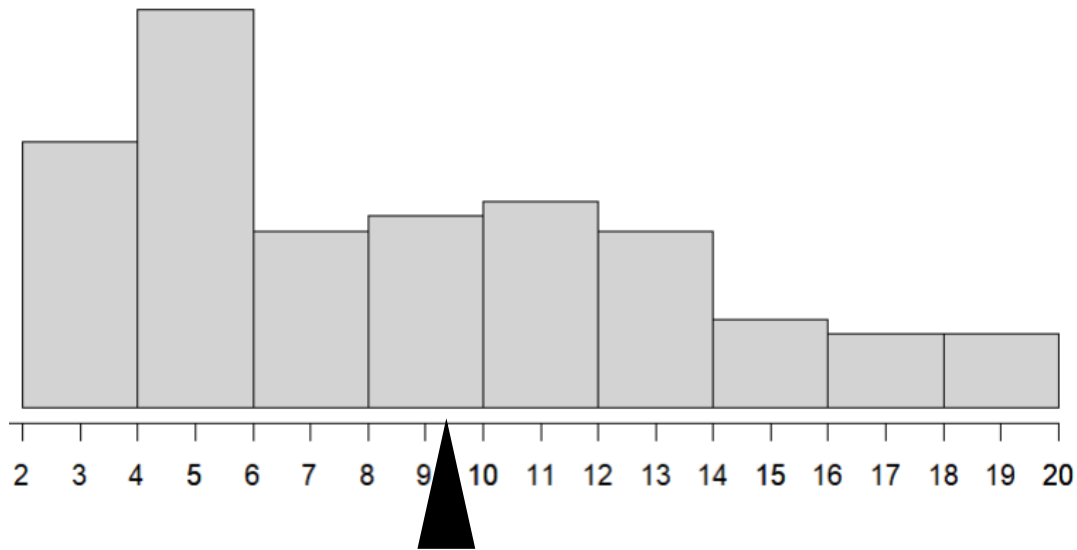
How to compute the median:

1. Compute the median by first ordering the observations from smallest value to largest value and choose the number in the middle
2. If the n is odd the median is the middle number
 - If n is even the median is the sum of the two middle values divided by 2

Practice: Calculate the Median

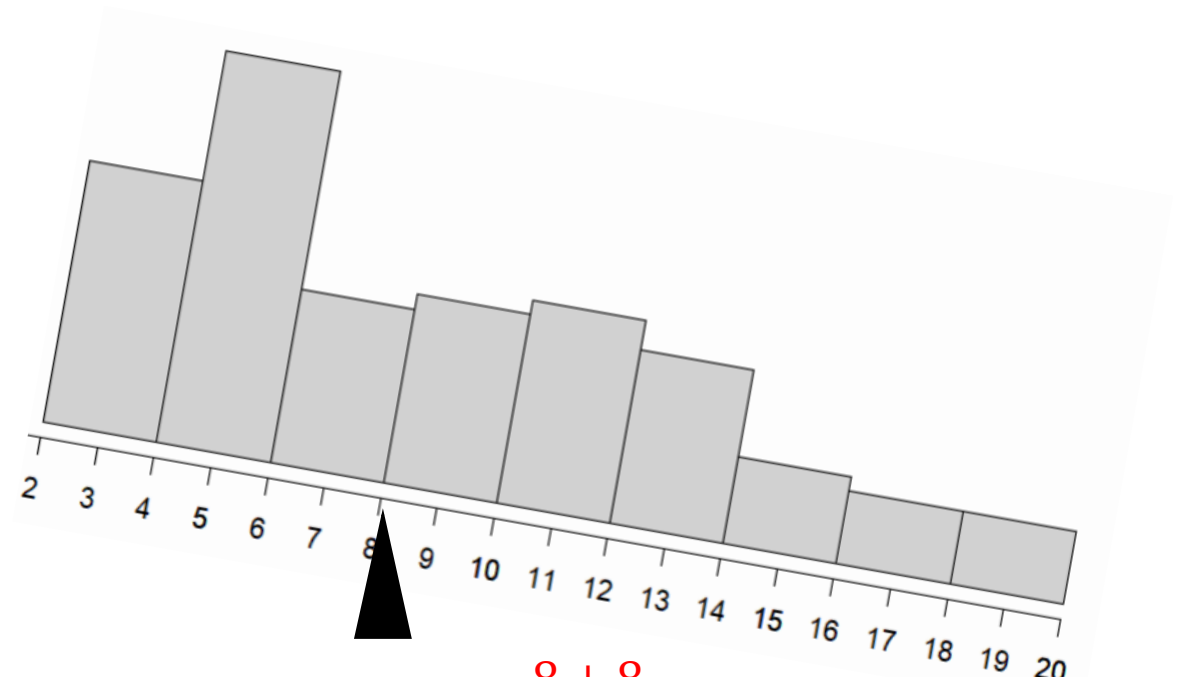
- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

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 12 12 12 12 12 12 12 12 13 13 13 14 14 14 14 14 14 14 15 15 15 16 16 16 16 17 17 17 18
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$$\bar{x} = 9.2$$

The mean is the center of gravity

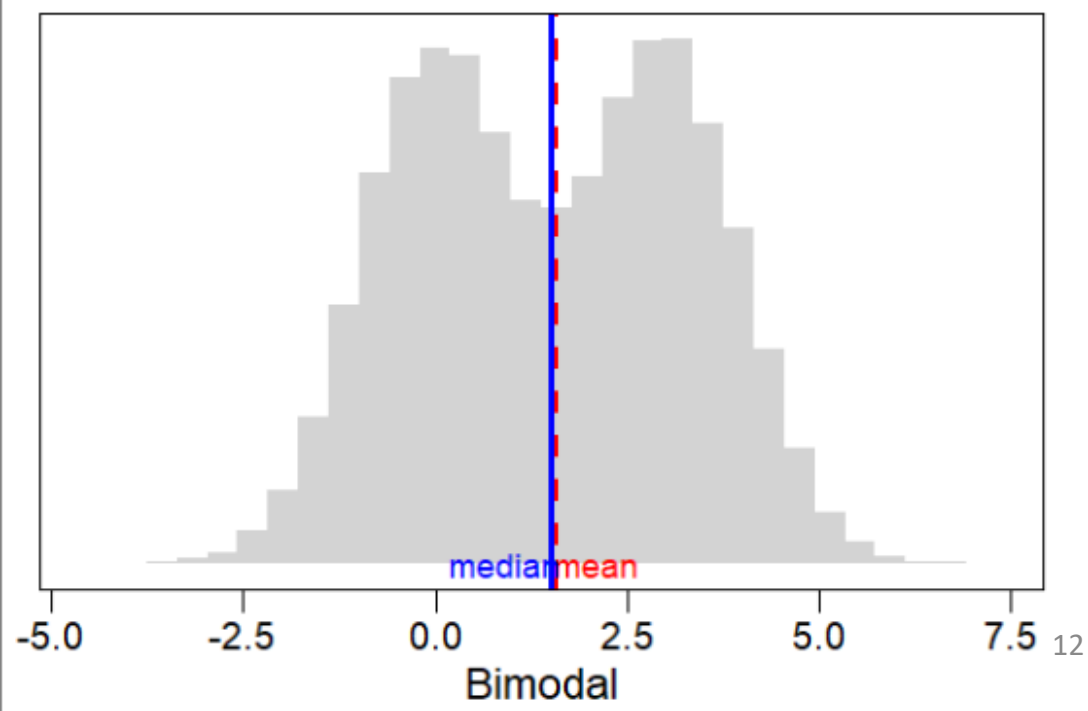
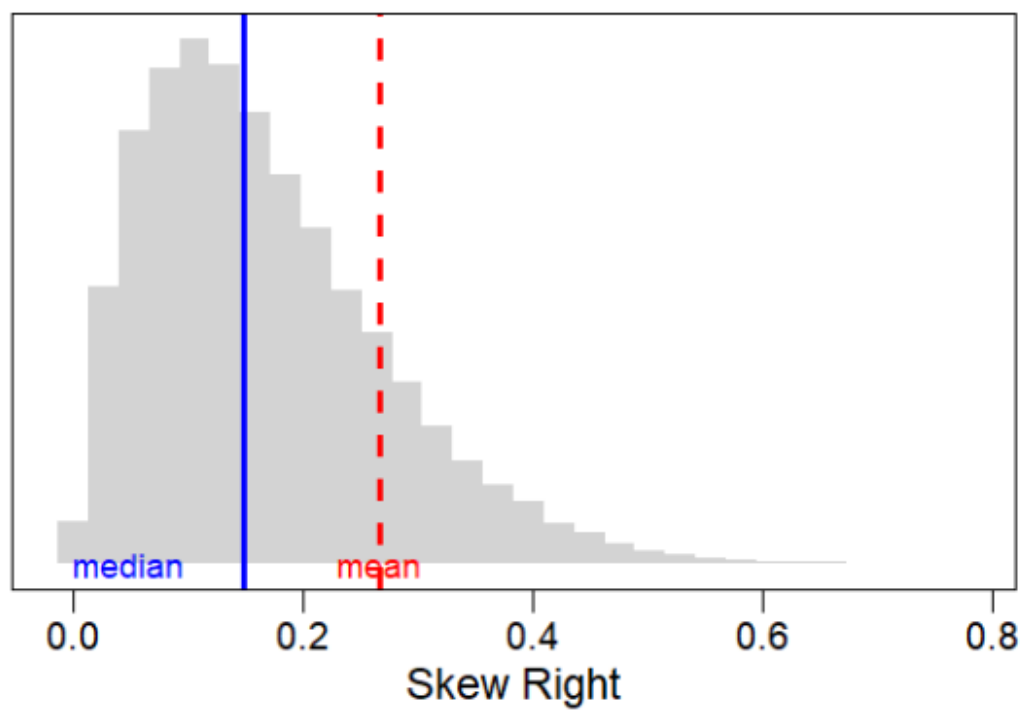
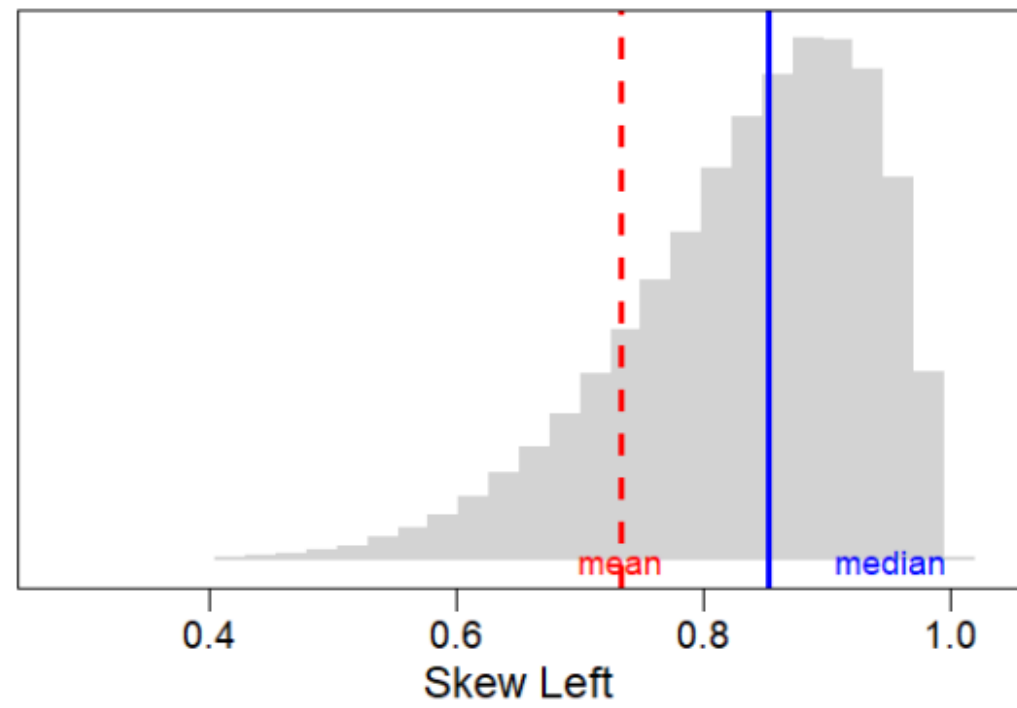
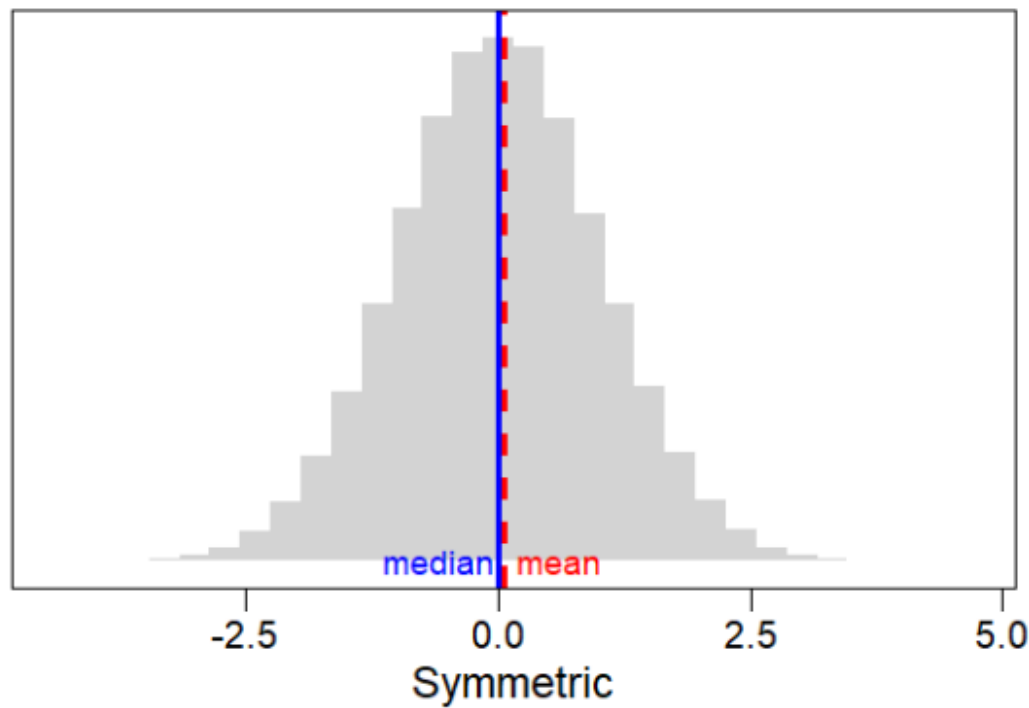


$$median = \frac{8 + 8}{2} = 8$$

The median is the middle value

Mean and median treat outliers differently

- $X = \{1, 3, 5, 5, 6, 7, 7, 8, 32\}$



The mode

- The **mode** is the value with the largest relative frequency (i.e the value that occurs most often)
 - Can be used with categorical data (mean and median cannot)
 - e.g the most frequent category
 - It may not be unique if two or more values have the same frequency
 - **Caution** for quantitative data, the mode may not anywhere near the center of the distribution.

Ex.)

Data = 1,1,4,5,6

Mode = 1

Data = 1,1,4,5,6,6

Mode 1, 6

Practice 2:

- Roll a six-sided die $n = 10$ times and record the number rolled each time

- Data = 1,2,3,3,4,4,4,5,6,6

x	$f(x)$	$rf(x)$
1	1	0.1
2	1	0.1
3	2	0.2
4	3	0.3
5	1	0.1
6	2	0.2

Compute the **mean** using all 3 equations:

Compute the **median**

Compute the **mode**

Mean, three ways:

$$\sum x / n = 38 / 10 = \mathbf{3.8}$$

$$(1/n) \sum x \cdot f(x) = (1 \cdot 1 + 2 \cdot 1 + 3 \cdot 2 + 4 \cdot 3 + 5 \cdot 1 + 6 \cdot 2) / 10 = 38 / 10 = \mathbf{3.8}$$

$$\sum x \cdot RF(x) = 1(.1) + 2(.1) + 3(.2) + 4(.3) + 5(.1) + 6(.2) = \mathbf{3.8}$$

Median:

ordered 1 2 3 3 4 4 4 5 6 6

n is even, so median = $(4 + 4) / 2 = \mathbf{4}$

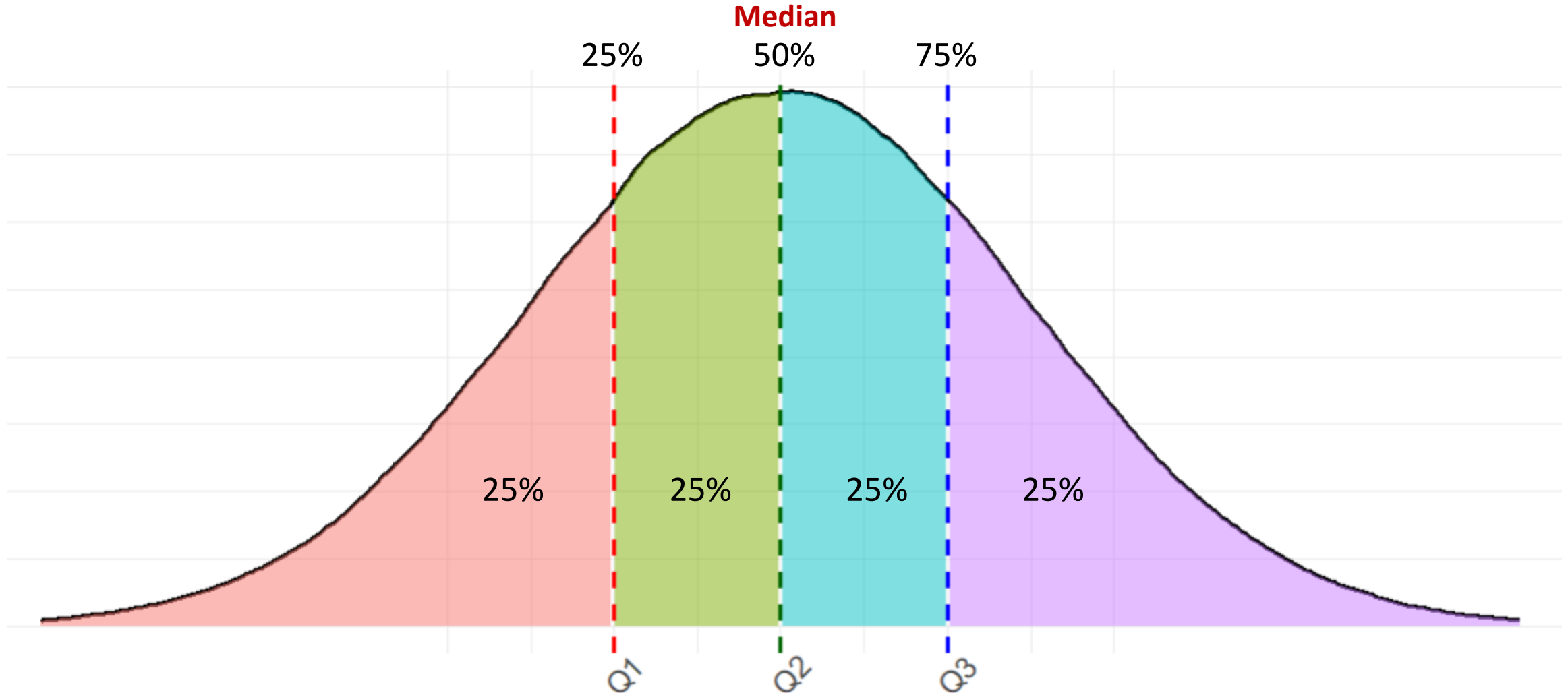
Mode:

4 appears three times, more than any other value $\rightarrow$ mode = **4**

Measures of Location: Quartiles and Percentiles

- The p^{th} **percentile** is a value such that p percent of the observations in a sample or population fall at or below that value
- Ex. The 50th percentile of any dataset is the median
- Three useful percentiles of a distribution are the **quartiles**
 1. The first **quartile Q1** is the 25th percentile of the data.
 2. The second **quartile Q2** is the 50th percentile or median of the data.
 3. The third **quartile Q3** is the 75th percentile of the data.
- The quartiles split a distribution in four equal parts each containing 25% of the observations

Measures of Position: Quartiles and Percentiles



Computing the Quartiles

- How to compute the quartiles:
 1. Arrange the data in increasing order
 2. Find the median and label as **Q2**
 3. Consider the lower half of the observations (excluding the median itself if n is an odd number).
 4. Mark the median for the lower half of the observations and label as **Q1**
 5. Consider the upper half of the observations (again excluding the median itself if n is odd)
 6. Mark the median for the upper half of the observations and label as **Q3**

Ex. Quartiles

- Consider the following data which come from $n = 20$ rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2, 2, 2, 2	2	3, 3, 3, 3, 4, 5, 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Quartiles from the cumulative relative frequency

- The crf column says what fraction of the data falls at or below each value
- So the percentiles can be read straight off the table
 - the p th percentile is the first value whose crf reaches $p/100$
- Q1: first crf at or above 0.25 gives $x = 3$
- Q2: first crf at or above 0.50 gives $x = 4$
- Q3: first crf at or above 0.75 gives $x = 5$
- The same Q1, Q2 and Q3 the halving method gave us

x	f(x)	rf(x)	crf(x)
1	1	0.1	0.1
2	1	0.1	0.2
3	2	0.2	0.4 ← Q1
4	3	0.3	0.7 ← Q2
5	1	0.1	0.8 ← Q3
6	2	0.2	1.0

One caution: when a crf lands exactly on 0.25, 0.50 or 0.75, the two methods can differ by one step. Keep using the halving method as the definition; read the table to check yourself.

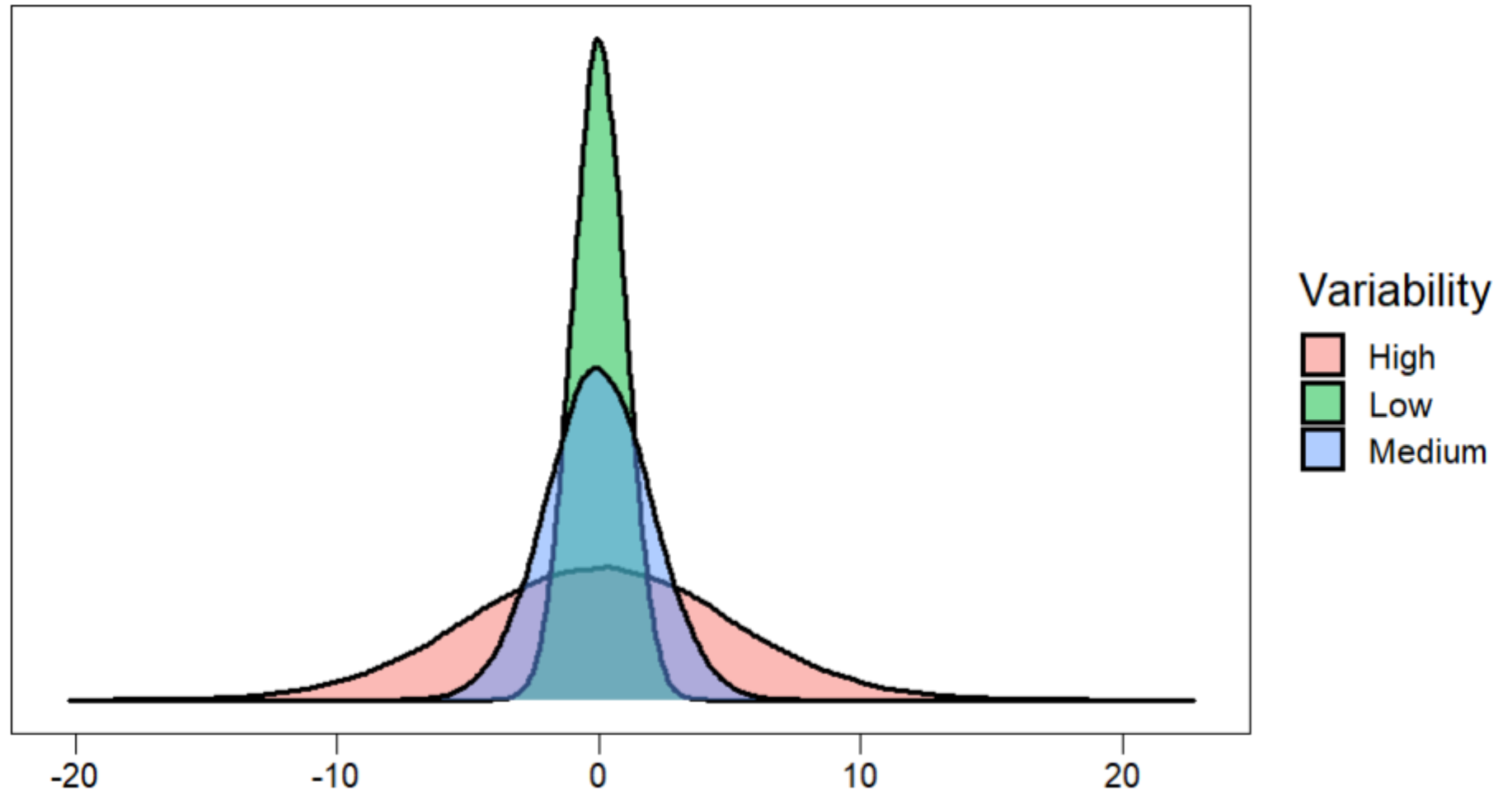
Practice: Quartiles from Exam Scores

Consider the following 15 exam scores of students in a statistics course

61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Compute the 3 quartiles Q_1 , Q_2 , and Q_3

Variability of A Distribution: Measures of Spread

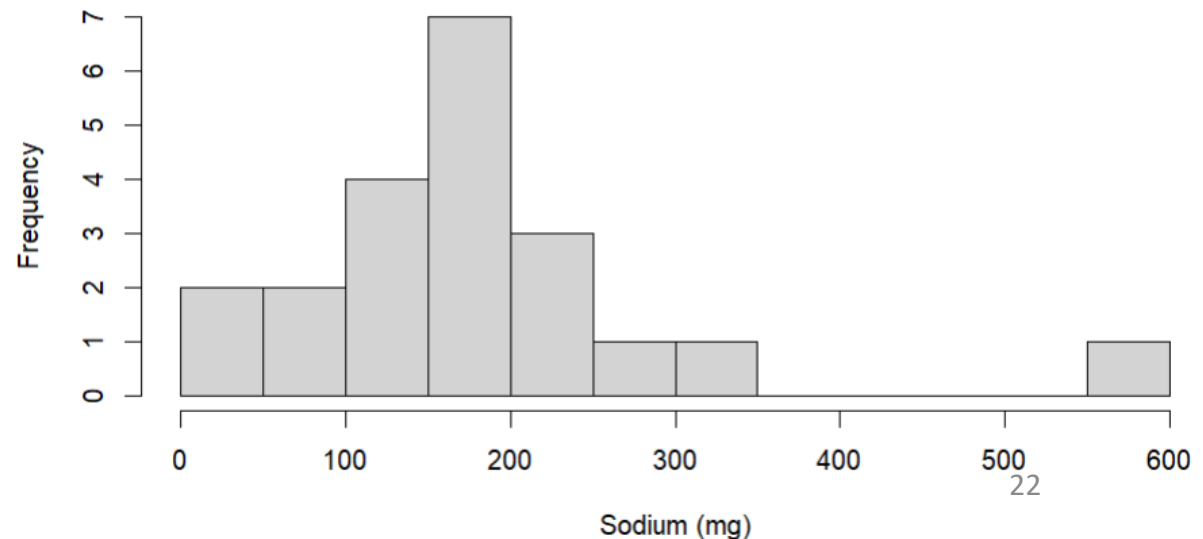
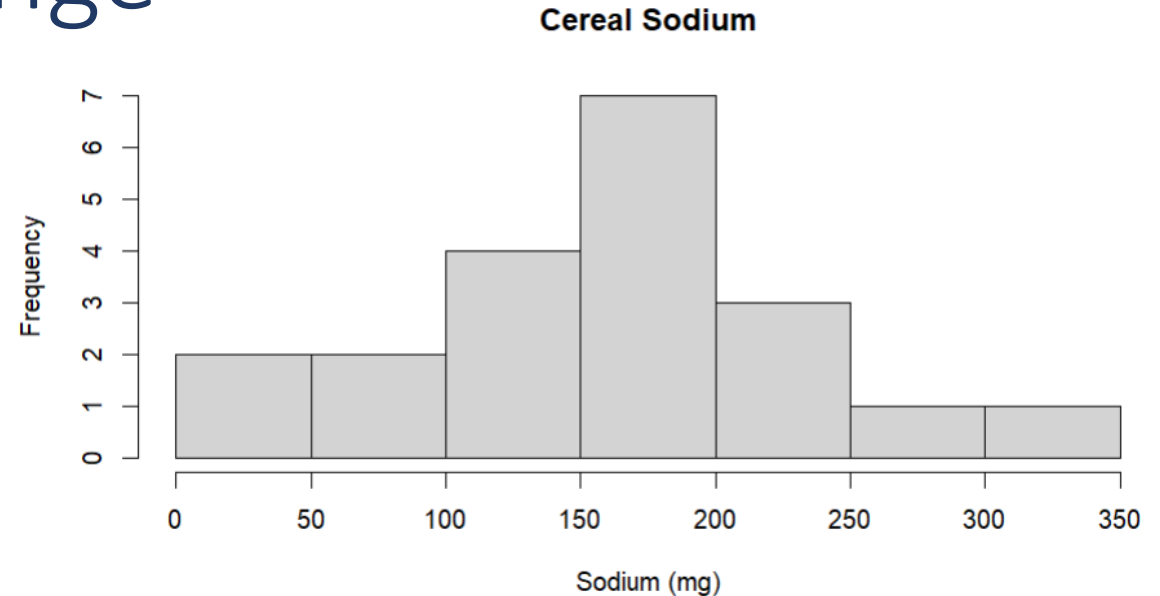


Measures of Spread: Range

- The **range** is a measure of the distance between the smallest and largest values in the data

The range can be computed with only two data points the minimum value and maximum value

- If the range of a set of data is large, then usually this indicates greater dispersion of values
- The range is severely affected by the presence of outliers
- We typically do not use the range to measure variability



Measures of Spread: Interquartile Range

- The **interquartile range (IQR)** measures the spread of the middle 50% of the observations
- It is resistant to outliers

$$IQR = Q3 - Q1$$

- The more variability the larger the value of the **IQR**
- **IQR** is a good choice for distributions that are highly skewed!

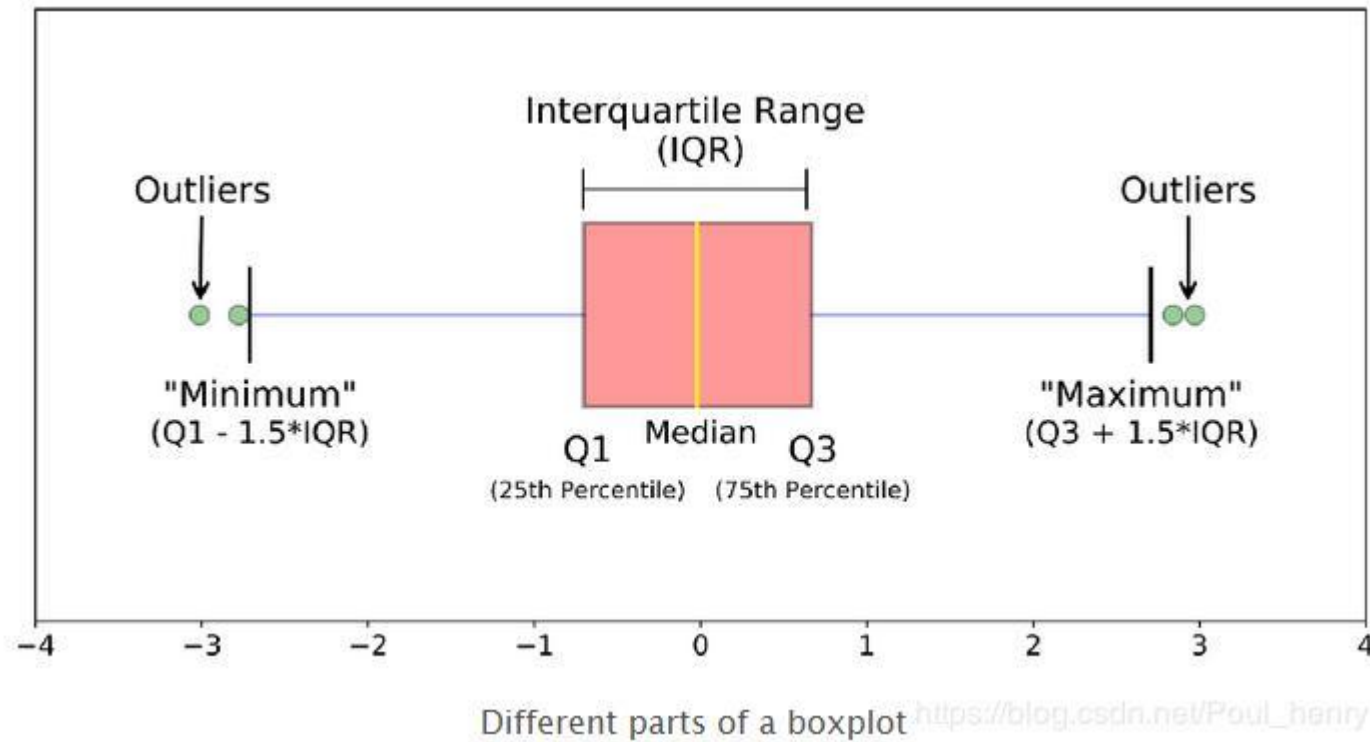
Practice: Finding quartiles and IQR

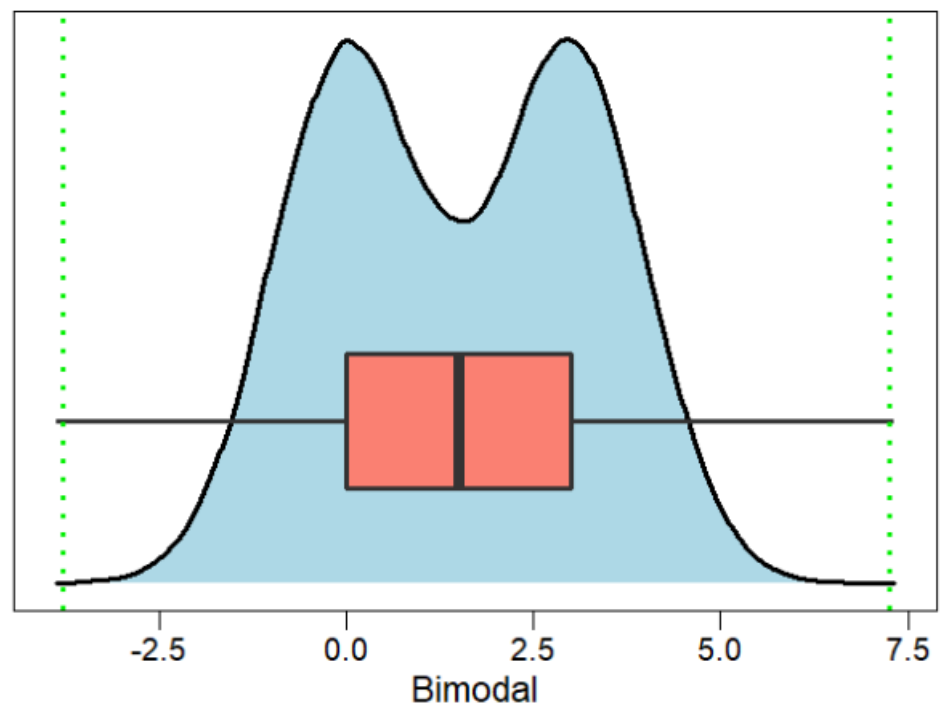
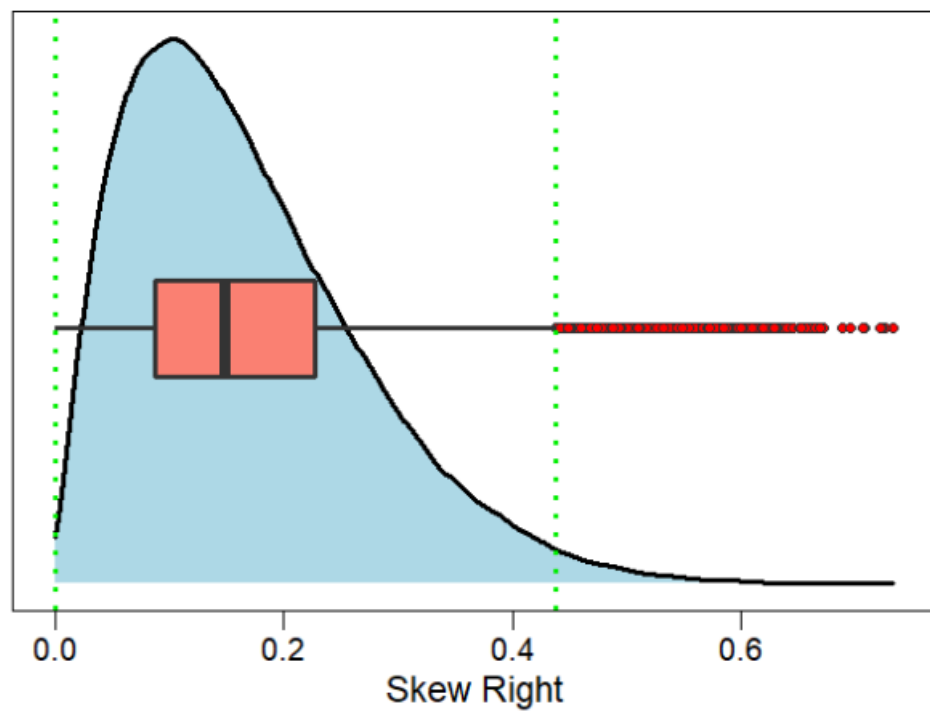
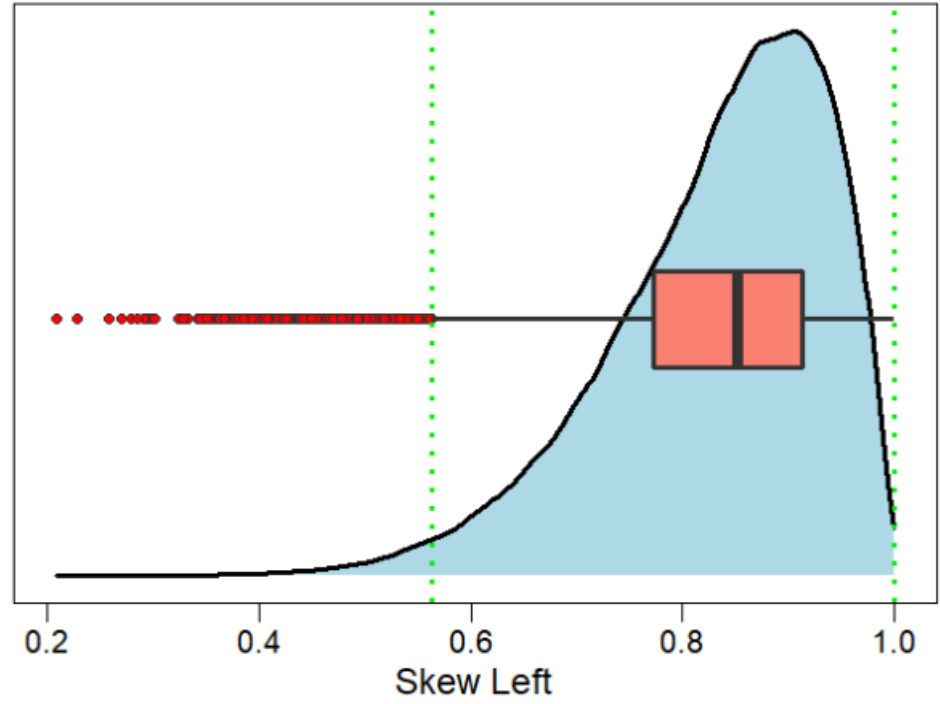
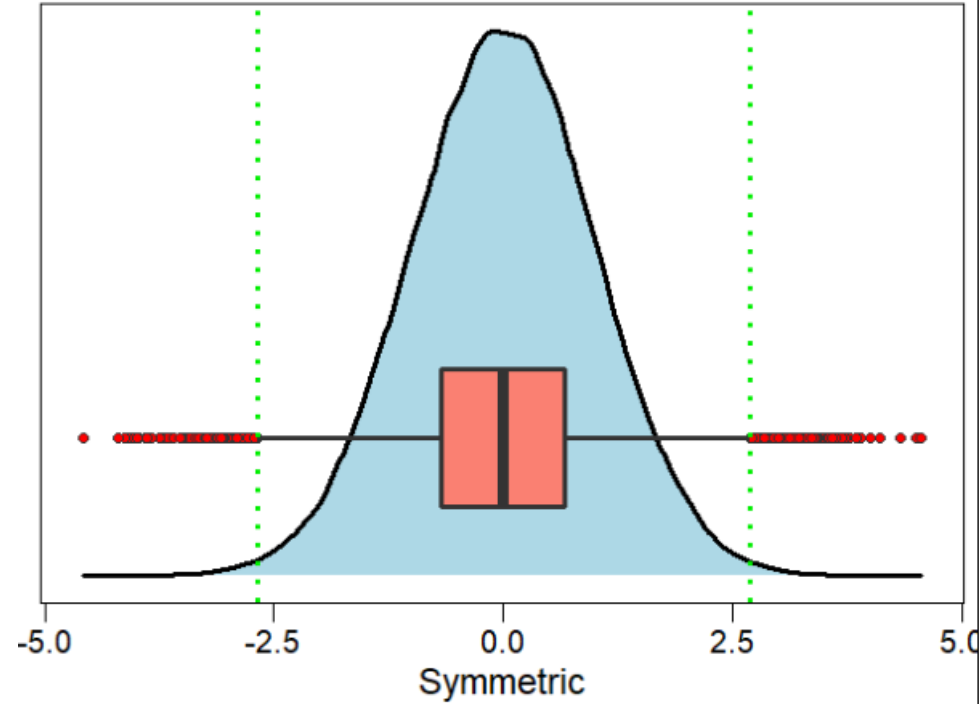
Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Compute the IQR

The Boxplot (Box and Whisker Plot): A five number summary

- Pros:
 - good for describing shape and location
 - Can be used to identify outliers
 - Length of whiskers indicates skew
 - Good for comparing two distributions or across categories
- Cons:
 - does not show certain features like mounds, or gaps as well as a histogram





Ex. Construct a Boxplot

- Consider the following data which come from 20 rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2 , 2, 2, 2, 2	3 , 3, 3, 3, 4, 5 , 5, 6, 6, 6	

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Practice: Construct a Boxplot

Construct the boxplot for student exam scores

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Ex2. Construct a Boxplot

Consider the following 12 observations of a quantitative variable X

$X = \{ -5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4 \}$

Compute the 5 number summary and draw a boxplot

Compare the IQR with the range

Measures of Spread: Deviation

- A better measure of variability that uses *all* the data is based on **deviations**
- **deviations** are the distances of each value from the mean of the data:

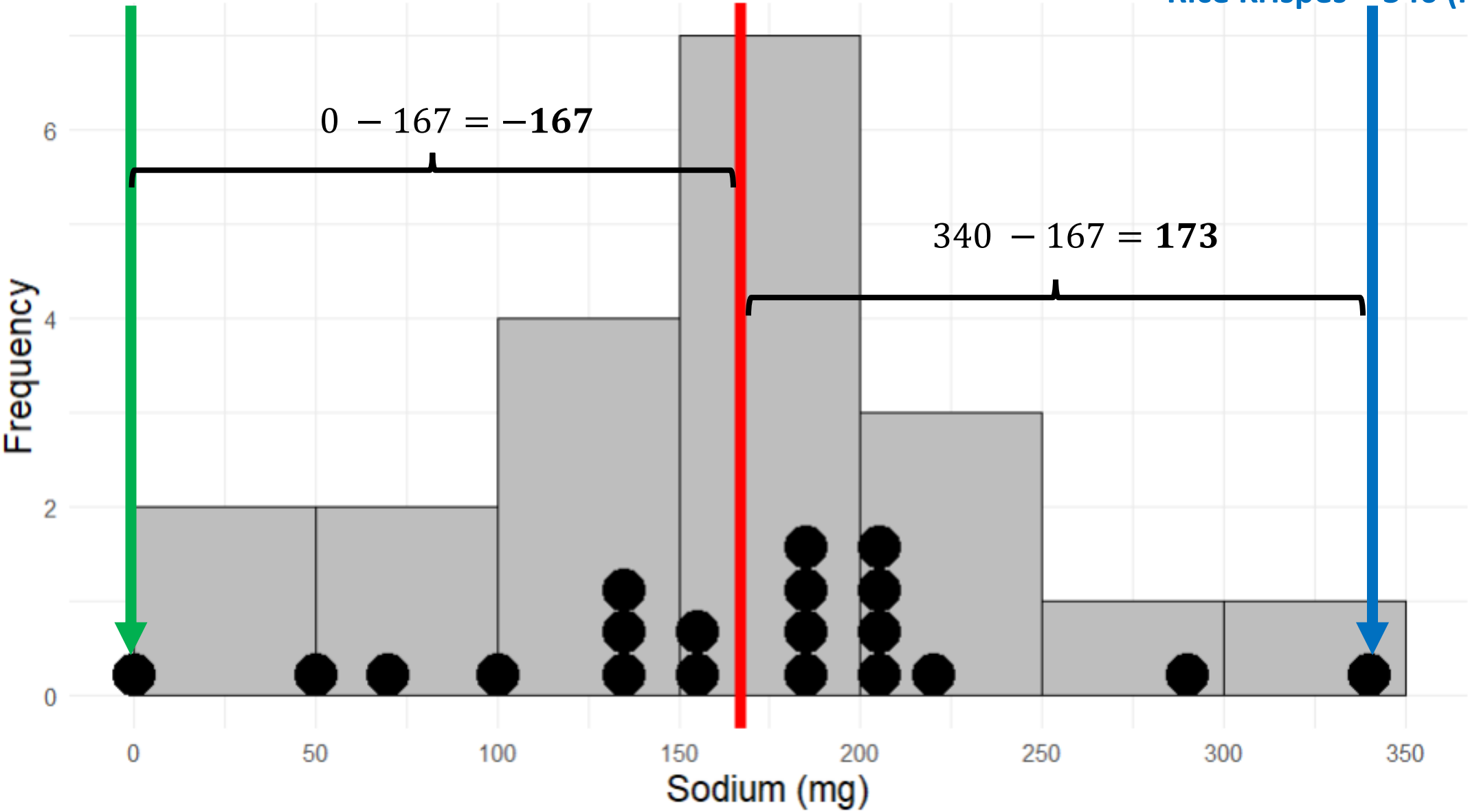
Deviation of an observation $x_i = (x_i - \bar{x})$

- Every observation will have a deviation from the mean

Frosted Mini Wheats = 0 (mg)

Mean = 167 (mg)

Rice Krispes = 340 (mg)



Measures of Spread: Variance

- The sum of all deviations is zero. $\sum_{i=1}^n (x_i - \bar{x}) = 0$
- We typically use either the **squared deviations** or their **absolute value**
Squared deviation of an observation $x_i = (x_i - \bar{x})^2$
- The **Variance** of a distribution is the average squared deviation from the mean

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sum $\sum_{i=1}^n (x_i - \bar{x})^2$ is called the sum of squares

Measures of Spread: Standard Deviation

- Since the variance uses the squared deviation, we usually take its square root called the **standard deviation**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- The standard deviation represents (roughly) the average distance of an observation from the mean
- The greater s is the greater the variability in the data is
- We denote the population parameter for the variance and standard deviation using σ for s and σ^2 for s^2

Why divide by $n - 1$?

- We divide by $n - 1$ because we have only $n - 1$ pieces of independent information for s^2
- Since the sum of the deviations must add to zero, then if we know the first $n - 1$ deviations we can always figure out the last one
- Ex.) suppose we have two data points and deviation of the first data point is $x - \bar{x} = -5$
 - Then the deviation of the second data point has to be 5 for the sum of deviations to be zero.

Try it out: Computing s and s^2

- Roll a six-sided die $n = 10$ times and record the number rolled each time
- Data = 1,2,3,3,4,4,4,5,6,6
- Mean = 3.8

